Logarithm
&
Its Applications

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Logarithm & its Applications

Meaning of Logarithm:

Logarithm is a rule which is used to solve complex algebraic expression.

Exponential Form:

Every positive real number can be expressed as $N = a^x$, $a > 0$, $a \neq 1$, $N>0$

$a \rightarrow$ base, $x \rightarrow$ exponent

Q.1 Express 16 using exponential form with base 2, base 4 and base 16.

Sol. (i) $N = 16$, $a_1 = 2$, Suppose x_1 = power

$2^{x_1} = 16 \Rightarrow 2^4 = 16$

(ii) $a_2 = 4$, power = x_2

$4^{x_2} = 16 \Rightarrow 4^2 = 16$

(iii) $a_3 = 16$, power = x_3

$16^{x_3} = 16 \Rightarrow 16^1 = 16$

Q.2 Express 81 using exponential form with base 3, base 9, and base 81.

Sol. (i) $N = 81$, $a_1 = 3$, x_1 = Power

$3^{x_1} = 81 \Rightarrow 3^4 = 81$

(ii) $a_2 = 9$, x_2 = Power

$9^{x_2} = 81 \Rightarrow 9^2 = 81$

(iii) $a_3 = 81$, x_3 = Power

$81^{x_3} = 81 \Rightarrow 81^1 = 81$

Q.3 Express 16 using exponential form with base '3'.

Sol. It means, $3^x = 16$

We know that, $3^2 = 9$ and $3^3 = 27$

$\Rightarrow 2 < x < 3 \Rightarrow x \in (2, 3)$

The types of 'x' can be find out using logarithm.

Note:

(i) What if a = 1.

Express 16 in exponential form with base a = 1

$1^x = 16 \Rightarrow$ there does not exist any 'x'.

For which $1^x = 16$

So we can not take base a = 1.

Ex.

Suppose a = –2 and $x = -\frac{1}{2}$ then

$(-2)^{-\frac{1}{2}}$ does not give real number.

(ii) What if a is negative.
Then for even root it will not be defined.
Hence for negative base it is not uniquely defined that with powers it will give real number or not.

Logarithmic form:

Logarithm of a number to some base is the exponent by which the base must be raised in order to get that number.

$\log_a N = x \Leftrightarrow a^x = N,\ N > 0,\ a > 0,\ a \neq 1$
$a \rightarrow$ base, $x \rightarrow$ exponent, $N \rightarrow$ number

Q.1 Find values of following:
(i) $\log_{10}1000$

Sol. Suppose $\log_{10}1000 = x \Leftrightarrow 10^x = 1000$
$\Rightarrow 10^x = 10^3$ (by comparing)
$\Rightarrow \mathbf{x = 3}$

Q. **(ii)** $\log_2 32$

Sol. Suppose $\log_2 32 = y \Leftrightarrow 2^y = 32$
$\Rightarrow 2^y = 2^5$ (by comparing)
$\Rightarrow \mathbf{y = 5}$

Q. **(iii)** $\log_5 625$

Sol. Suppose $\log_5 625 = x \Leftrightarrow 5^x = 625$
$\Rightarrow 5^x = 5^4$ (by comparing)
$\Rightarrow \mathbf{x = 4}$

Q. **(iv)** $\log_{\sqrt{2}} 64$

Sol. Suppose $\log_{\sqrt{2}} 64 = y \Leftrightarrow \left(\sqrt{2}\right)^y = 64$

$\Rightarrow 2^{\frac{y}{2}} = 2^6$

$\Rightarrow \frac{y}{2} = 6$ (by comparing)

$\Rightarrow \mathbf{y = 12}$

Q. **(v)** $\log_{2\sqrt{3}} 1728$

Sol. Suppose $\log_{2\sqrt{3}} 1728 = x \Leftrightarrow \left(2\sqrt{3}\right)^x = 1728 = 12^3$

$$\Rightarrow \left(2\sqrt{3}\right)^x = \left(\left(2\sqrt{3}\right)^2\right)^3$$

$$\Rightarrow \left(2\sqrt{3}\right)^x = \left(2\sqrt{3}\right)^6 \text{ (by comparing)}$$

$$\Rightarrow \mathbf{x = 6}$$

Q. **(vi)** $\log_{16} 32$

Sol. Suppose $\log_{16} 32 = y \Leftrightarrow 16^y = 32$

$$\Rightarrow (2^4)^y = 2^5$$

$$\Rightarrow 2^{4y} = 2^5$$

$$\Rightarrow 4y = 5 \text{ (by comparing)}$$

$$\Rightarrow \mathbf{y = \frac{5}{4}}$$

Q. **(vii)** $\log_{\sqrt[3]{7}} 2401$

Sol. Suppose $\log_{\sqrt[3]{7}} 2401 = x \Leftrightarrow \left(\sqrt[3]{7}\right)^x = 2401$

$$\Rightarrow 7^{\frac{x}{3}} = 7^4$$

$$\Rightarrow \frac{x}{3} = 4 \text{ (by comparing)}$$

$$\Rightarrow \mathbf{x = 12}$$

Q. **(viii)** $\log_3\left(\log_3(27)^3\right)$

Sol. First we will find the value of $\log_3(27)^3 = y$

$$\Rightarrow \log_3(3^3)^3 = y$$

$$\Rightarrow \log_3 3^9 = y$$

$$\Rightarrow y = 9$$

Then it reduces to $\log_3 9$ and we know that $\log_3 9 = 2$

Hence, $\log_3(\log_3(27)^3) = \mathbf{2}$

Q.2 Find 'x' for which $\log_{(x-1)}(2x + 1)$ is defined.

(A) $x \in (1, \infty)$

(B) $x \in \left(\frac{-1}{2}, \infty\right) - \{2\}$

(C) $x \in (1, \infty) - \{2\}$

(D) $x \in \left(\frac{-1}{2}, 1\right)$

Sol. **(C)**

We will check when $\log_{(x-1)}(2x+1)$ is defined.

$x - 1 > 0$ and $(x - 1) \neq 1$ and $(2x + 1) > 0$

$\Rightarrow x > 1$ and $x \neq 2$ and $x > \frac{-1}{2}$

$\Rightarrow x \in (1, \infty)$ and $x \neq 2$ and $x \in \left(\frac{-1}{2}, \infty\right)$

Intersection gives $x \in (1, \infty)$ and $x \neq 2 \Rightarrow x \in (1, \infty) - \{2\}$

It can also be written as

$\mathbf{x \in (1, 2) \cup (2, \infty)}$

Q.3 Find complete set of real values of 'x' for which $\log_{(2x-3)}(x^2-5x-6)$ is defined.

Sol. $2x - 3 > 0$ and $(2x - 3) \neq 1$ and $(x^2 - 5x - 6) > 0$

$\Rightarrow x > \frac{3}{2}$ and $x \neq 2$ and $(x - 6)(x + 1) > 0$

$\Rightarrow x > \frac{3}{2}$ and $x \neq 2$ and $x \in (-\infty, -1) \cup (6, \infty)$

$\Rightarrow x \in \left(\frac{3}{2}, \infty\right)$ and $x \neq 2$ and $x \in (-\infty, -1) \cup (6, \infty)$

Intersection gives, $x \in (6, \infty)$

Q.4 Let $a = \log_{1/2}16$

$b = \log_3(\tan 30°)$

$c = \log_{2-\sqrt{3}}\left(2+\sqrt{3}\right)$

$d = \log_2(\log_2 4)$

then $a + b + c + d$ is

(A) $-\frac{7}{2}$ (B) $-\frac{9}{2}$ (C) $-\frac{5}{2}$ (D) -4

Sol. **(B)**

$a = \log_{1/2}16$

$\Rightarrow \left(\frac{1}{2}\right)^a = 16$

$\Rightarrow 2^{-a} = 2^4$

$\Rightarrow a = -4$

$b = \log_3(\tan 30°)$

$\Rightarrow 3^b = \tan 30° = \frac{1}{\sqrt{3}} = 3^{-\frac{1}{2}}$

$\Rightarrow b = -\frac{1}{2}$

$$c = \log_{2-\sqrt{3}}\left(2+\sqrt{3}\right)$$

$$\Rightarrow \left(2-\sqrt{3}\right)^{c} = \frac{\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)}{\left(2-\sqrt{3}\right)}$$

$$\Rightarrow \left(2-\sqrt{3}\right)^{c} = (4-3)\left(2-\sqrt{3}\right)^{-1}$$

$c = -1$

$d = \log_2(\log_2 4)$

we know that $\log_2 4 = 2$

so, $d = \log_2 2 \Rightarrow d = 1$

then $a+b+c+d = -4-\frac{1}{2}-1+1 = -\frac{9}{2}$

Important Deduction:

(i) $\log_N N = 1$

(ii) $\log_N 1 = 0$

(iii) $\log_{1/N} N = -1$

(i) Proof : Suppose $\log_N N = x \Leftrightarrow N^x = N^1 \Rightarrow x = 1$

(ii) Proof : Suppose $\log_N 1 = y \Leftrightarrow N^y = 1 = N^0 \Rightarrow y = 0$

(iii) Proof : Suppose $\log_{\frac{1}{N}} N = z \Leftrightarrow \left(\frac{1}{N}\right)^z = N \Rightarrow N^{-z} = N^1 \Rightarrow z = -1$

Q.5 Find values of following

(i) $\log_{\sin 30^\circ}\cos 60^\circ$

Sol. First simplify this, we get $\log_{\frac{1}{2}}\left(\frac{1}{2}\right)$

so $\log_{\frac{1}{2}}\left(\frac{1}{2}\right) = 1$

Know the facts

We know that $\log_N N = 1$

Q. **(ii)** $\log_{4/3} 1.\overline{3}$

Sol. Let $N = 1.\overline{3} = 1.3333....$

$N = 1.3333...$ ...(i)

$10\ N = 13.33333.....$ ...(ii)

Know the facts

We know that $\log_N N = 1$

Subtracting (i) from (ii), we get

$-9N = -12$

$N = \frac{-12}{-9} = \frac{4}{3}$

We get $\log_{\frac{4}{3}}\left(\frac{4}{3}\right) = 1$

Q. **(iii)** $\log_5 \sqrt{5\sqrt{5\sqrt{5}\ldots}}$

Sol. **Sol.** $N = \sqrt{\underbrace{5\sqrt{5\sqrt{5}\ldots}}_{N}}$

$N = \sqrt{5N}$

$N^2 = 5N$

$\Rightarrow N^2 - 5N = 0$

$\Rightarrow N(N-5) = 0$

$\Rightarrow N = 0$ or $N = 5$ but $N = 0$ is not possible

We get $\log_5 5 = \mathbf{1}$

Q. **(iv)** $\log_2(\sin^2 x + \cos^2 x)$

Sol. We know that $\sin^2 x + \cos^2 x = 1$

So we get , $\log_2 1 = \mathbf{0}$

$\because \log_N 1 = 0$

Q. **(v)** $\log(\tan 1°).\log(\tan 2°).\log(\tan 3°)\ldots\log(\tan 89°)$

Sol. $\log(\tan 1°).\ \log(\tan 2°)\ldots\ \log(\tan 45°)\ldots\log(\tan 89°)$

$= \log(\tan 1°).\ \log(\tan 2°)\ldots\ \log 1\ldots\log(\tan 89°)$

$= \mathbf{0}$

Q. **(vi)** $\log(\tan 1°.\tan 2°.\tan 3°\ldots\ \tan 89°)$

Sol. We get

$\log(\tan 1°\cdot\tan 2°\cdot\tan 3°\ldots\tan 43°\cdot\tan 44°\cdot\tan 45°\cdot\tan 46°\ldots\tan 89°)$

$= \log(\tan 1°\cdot\tan 2°\ldots\ \tan 44°\cdot\tan 45°\cdot\tan(99°-44°)\ldots\tan(90-1)°)$

$= \log(\tan 1°\cdot\tan 2°\ldots\ \tan 44°\cdot 1\cdot\cot 44°\cdot\cot 43°\ldots\cot 1°)$

$= \log(\tan 1°\cdot\cot 1°\cdot\tan 2°\cot 2°\ldots\tan 44°\cot 44°)$

$= \log 1 = \mathbf{0}$

Using $\tan(90°-\theta) = \cot\theta$ and $\tan\theta.\cot\theta = 1$

Q. **(vii)** $\log_2(\log_3(\log_3(27)^3))$

Sol. Let's assume
$\log_3(27)^3 = y$
$\Rightarrow \log_3(3^3)^3 = y$
$\Rightarrow \log_3 3^9 = y$
$\Rightarrow y = 9$
Then it reduces to $\log_2 \underbrace{(\log_3 9)}_{2} = \log_2 2 = 1$

Q. **(viii)** $\dfrac{(\log_{100}10)\left(\log_2(\log_4 2)\right)\log_4\left(\log_2^2(256)^2\right)}{\log_4 8 + \log_8 4}$

Sol.

Let $a = \log_{100}10$
$\Rightarrow 100^a = 10$
$\Rightarrow (10)^{2a} = 10^1$
$\Rightarrow 2a = 1$
$\Rightarrow a = \dfrac{1}{2}$

$b = \log_2(\log_4 2)$
$\Rightarrow b = \log_2 \dfrac{1}{2}$
$\Rightarrow 2^b = 2^{-1}$
$\Rightarrow b = -1$

Let $x = \log_2^2(256)^2 \Rightarrow x = \left[\log_2(2^8)^2\right]^2 \Rightarrow x = \left(\log_2 2^{16}\right)^2$

$\Rightarrow x = 16^2 = 256$

$c = \log_4\left(\log_2^2(256)^2\right)$
So we get $c = \log_4 256$
$\Rightarrow 4^c = 4^4$
$\Rightarrow c = 4$

$d = \log_4 8$
$\Rightarrow 4^d = 8$
$\Rightarrow (2^2)^d = 2^3$
$\Rightarrow 2^{2d} = 2^3$
$\Rightarrow 2d = 3$
$\Rightarrow d = \dfrac{3}{2}$

$e = \log_8 4$
$\Rightarrow 8^e = 4$
$\Rightarrow (2^3)^e = 2^2$
$\Rightarrow 3e = 2$
$\Rightarrow e = \dfrac{2}{3}$

Then given expansion reduces to $\dfrac{abc}{d+e} = \dfrac{\left(\frac{1}{2}\right)(-1)(4)}{\frac{3}{2}+\frac{2}{3}} = \dfrac{(-2)6}{(9+4)} = -\dfrac{12}{13}$

Q.6 If $\log_2(\log_2(\log_3 x)) = 0 = \log_2(\log_3(\log_2 y))$ then find $x + y$.

Sol. $\log_2(\log_2(\log_3 x) = 0 \Leftrightarrow \log_2(\log_3 x) = 2^0 = 1$ (Change it into exponential form)
$\Leftrightarrow \log_3 x = 2^1 = 2$
$\Leftrightarrow x = 3^2 = 9$
$x = 9$

Similarly, $\log_2(\log_3(\log_2 y) = 0 \Leftrightarrow \log_3(\log_2 y) = 2^0 = 1$
(Change it into exponential form)
$\Leftrightarrow \log_2 y = 3^1 = 3$
$\Leftrightarrow y = 2^3 = 8$

Then $x + y = 9 + 8 = \mathbf{17}$

Fundamental Identify:

$$a^{\log_a N} = N$$

Proof:

Let $\log_a N = x$ then it changes into
$a^x = N \Leftrightarrow \log_a N = x$
So we get $a^{\log_a N} = N$

Q. Find the value of following: $3^{\log_3 10}$

Sol. Using Identify, $a^{\log_a N} = N$
We get $3^{\log_3 10} = 10$

Properties of logarithm:

If m, n are positive real numbers, $a > 0$, $a \neq 1$ then

(i) $\log_a mn = \log_a m + \log_a n$

Proof:

Suppose $\log_a m = x$ and $\log_a n = y$
$a^x = m$ and $a^y = n$
then $mn = a^x.a^y = a^{x+y}$
again change it into logarithmic form, $x + y = \log_a mn$
we get $\log_a mn = \log_a m + \log_a n$

Q.1 Solve : $\log_{10}2 + \log_{10}5$

Sol. $\log_{10}2 + \log_{10}5$
$= \log_{10}(2\times5)$ (using P(1))
$= \log_{10}10$
$= \mathbf{1}$

Note:

General version:

Suppose $a > 0, a \neq 1, N_1, N_2, N_3, \ldots, N_r > 0$

$\log_a(N_1.N_2.N_3 \ldots N_r) = \log_a N_1 + \log_a N_2 + \log_a N_3 + \ldots + \log_a N_r$

(ii) $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$

Proof:

Given $a > 0, a \neq 1, m > 0, n > 0$

Then suppose $\log_a m = x$ & $\log_a n = y$

$\Rightarrow \quad a^x = m$ & $a^y = n$

$\Rightarrow \quad \frac{m}{n} = \frac{a^x}{a^y} = a^{x-y}$

Change it into logarithmic form,

$x - y = \log_a\left(\frac{m}{n}\right)$

$\Rightarrow \log_a m - \log_a n = \log_a\left(\frac{m}{n}\right)$

Q.2 Solve : $\log_2 10 - \log_2 5$

Sol. $\log_2 10 - \log_2 5 = \log_2 \frac{10}{5}$ (using P(2))

$= \log_2 2$

$= 1$

(iii) $\log_a m^n = n \log_a m$

Proof:

Let's say $\log_a m = \alpha \Leftrightarrow a^\alpha = m$

then $m^n = (a^\alpha)^n = a^{n\alpha}$

Change it into logarithmic form, then

$n\alpha = \log_a m^n$

$n\log_a m = \log_a m^n$

Q.3 **Q3.** Solve $\log_2 32$

Sol. $\log_2 32 = \log_2 2^5$
$= 5\log_2 2$
$= \mathbf{5}$

Q.4 Find values of following :
(i) $\log_{385} 5 + \log_{385} 7 + \log_{385} 11$

Sol. Using property $\log_a m + \log_a n + \log_a p = \log_a mnp$
We get $\log_{385} 5.7.11$
$= \log_{385} 385$
$= \mathbf{1}$

Q. **(ii)** $\log_4(\log_2(\log_5 625))$

Sol. $\log_4(\log_2(\log_5 5^4))$
$= \log_4(\log_2(4\log_5 5))$
$= \log_4(\log_2(2^2 \times 1))$
$= \log_4(2\log_2 2)$
$= \log_4 2$
$= \mathbf{1/2}$

Q. **(iii)** $\log_{11}(11)^{\log_{11} 1331}$

Sol.
$\log_{11}(11)^{\log_{11} 1331}$
$= \log_{11} 11^{\log_{11} 11^3}$ [Using $\log_a m^n = n\log_a m$]
$= \log_{11} 11^{3\log_{11} 11}$
$= \log_{11} 11^3$
$= 3\log_{11} 11$
$= \mathbf{3}$

Q. **(iv)** $\log_2 10 - \log_2 5 + \log_2 8$

Sol. $\log_2 10 - \log_2 5 + \log_2 8$
$= \log_2 \frac{10}{5} + \log_2 8$ [Using P(1)]
$= \log_2 2 + \log_2 2^3$
$= 1 + 3\log_2 2$
$= 1 + 3$
$= \mathbf{4}$

Q. (v) $\log_2[\log_4(\log_{10}16^4 + \log_{10}25^8)]$

Sol. $\log_2[\log_4(\log_{10}16^4 + \log_{10}25^8)]$

$= \log_2[\log_4(\log_{10}16^4.25^8)]$

$= \log_2[\log_4(\log_{10}2^{4\times4}.5^{2\times8})]$

$= \log_2[\log_4(\log_{10}(10)^{16})]$

$= \log_2[\log_4(16\log_{10}10)]$

$= \log_2[\log_4 4^2]$

$= \log_2 2(\log_4 4)$

$= \log_2 2 = \mathbf{1}$

Q.5 $\sum_{n=1}^{1023} \log_2\left(1+\frac{1}{n}\right)$ is equal to

(A) 8 (B) 9 (C) 10 (D) 12

Sol. **(C)**

$\sum_{n=1}^{1023} \log_2\left(\frac{n+1}{n}\right)$

$= \log_2\left(\frac{2}{1}\right) + \log_2\left(\frac{3}{2}\right) + \log_3\left(\frac{4}{3}\right) + \ldots + \log_2\left(\frac{1024}{1023}\right)$

$= \log_2\left(\frac{2}{1}.\frac{3}{2}.\frac{4}{3}\ldots\frac{1024}{1023}\right)$

$= \log_2 1024$

$= \log_2 2^{10}$

$= 10\log_2 2 = \mathbf{10}$

Q.6 Find the value of $\log_{10}2 + 16\log_{10}\left(\frac{16}{15}\right) + 12\log_{10}\left(\frac{25}{24}\right) + 7\log_{10}\left(\frac{81}{80}\right)$.

Sol. $\log_{10}2 + 16\log_{10}16 - \log_{10}15 + 12\log_{10}25 - 12\log_{10}24 + 7\log_{10}81 - 7\log_{10}80$

$= \log_{10}2 + 16\log_{10}2^4 - 16\log_{10}(3\times5) - \log_{10}(3\times5) + 12\log_{10}5^2 - 12\log_{10}(2^3.3)$
$+ 7\log_{10}3^4 - 7\log_{10}(2^4.5$

$= \log_{10}2 + 64\log_{10}2 - \log_{10}3 - \log_3 5 + 24\log_{10}5 - 36\log_{10}2$
$- 12\log_{10}3 + 28\log_{10}3 - 28\log_{10}2 - 7\log_{10}5$

$= (1 + 64 - 36 - 28)\log_{10}2 + (-16 - 12 + 28)\log_{10}3 + (-16 - 24 - 7)\log_{10}5$

$= \log_{10}2 + \log_{10}5$

$= \log_{10}10 = \mathbf{1}$

Base Changing theorem:

$$\log_b a = \frac{\log_c a}{\log_c b}, a > 0, b > 0, c > 0, a \neq 1, b \neq 1, c \neq 1$$

Proof:

Suppose $\log_b a = x \Leftrightarrow a = b^x$

$\log_c a = \log_c b^x$

$\log_c a = x \log_c b$ **(By taking logarithm)**

$\frac{\log_c a}{\log_c b} = x = \log_b a$

$\Rightarrow \log_b a = \frac{\log_c a}{\log_c b}$

Note:

$$\log_a b = \frac{1}{\log_b a}$$

Proof:

$\frac{\log_c b}{\log_c a} = \frac{1}{\frac{\log_c a}{\log_c b}}$ **(By using base Changing Theorem)**

$\frac{\log_c b}{\log_c a} = \frac{\log_c b}{\log_c a}$ **(Both are equal to each other)**

Hence proved.

Q.7 Prove the following: $\log_b a \cdot \log_c b \cdot \log_d c = \log_d a$

Sol. $\log_b a \cdot \log_c b \cdot \log_d c = \frac{\log a}{\log b} \cdot \frac{\log b}{\log c} \cdot \frac{\log c}{\log d} = \frac{\log a}{\log d}$

$= \log_d a$ **(Using Base-Changing Theorem)**

Q.8 If $\log_2 3 . \log_3 4 . \log_4 5 ... \log_n (n+1) = 10$. Find 'n'.

Sol. $\log_2 3 . \log_3 4 . \log_4 5 ... \log_n (n+1) = 10$

$$\frac{\log 3}{\log 2} \cdot \frac{\log 4}{\log 3} \cdot \frac{\log 5}{\log 4} \cdot .... \cdot \frac{\log(n+1)}{\log n} = 10$$

$$\Rightarrow \frac{\log(n+1)}{\log 2} = 10$$ **(Using Base-Changing Theorem)**

$\Rightarrow \log_2(n+1) = 10$

$\Rightarrow 2^{10} = n + 1$

$\Rightarrow n = 1024 - 1 \Rightarrow$ **n = 1023**

Property of logarithm:

$$a^{\log_b c} = c^{\log_b a}$$

Proof:

$$a^{\log_b c} = a^{\frac{\log_a c}{\log_a b}}$$

$$= \left(a^{\log_a c}\right)^{\frac{1}{\log_a b}}$$ **(Using Base-Changing theorem)**

$$= (c)^{\frac{1}{\log_a b}} = c^{\log_b a}$$

Hence $a^{\log_b c} = c^{\log_b a}$

Property of logarithm:

$$\log_{a^k}\left(m^n\right) = \frac{n}{k}\log_a m$$ wherever defined

Proof:

$$\text{L.H.S.} = \frac{\log_c m^n}{\log_c a^k} = \frac{n\log_c m}{k\log_c a} = \frac{n}{k}\log_a m = \text{R.H.S.}$$ **(Using Base-changing theorem)**

Hence Proved.

Q.9 The value of $7^{\log_3 5} + 3^{\log_5 7} - 5^{\log_3 7} - 7^{\log_5 3}$ is equal to

(A) 3 (B) 5 (C) 7 (D) 0

Sol. **(D)**

$$7^{\log_3 5} + 7^{\log_5 3} - 7^{\log_3 5} - 7^{\log_5 3} = 0$$ **(Using property $a^{\log_c b} = b^{\log_c a}$)**

Q.10 Let $x_1 = \log_{16}32$, $x_2 = \log_{625}125$, $x_3 = \log_8 16$, $x_4 = \log_{\sqrt{2}}8$. Find $\sum_{i=1}^{4} x_i$.

Sol. $x_1 = \log_{2^4}2^5 = \frac{5}{4}\log_2 2 = \frac{5}{4}$

$x_2 = \log_{5^4}5^3 = \frac{3}{4}\log_5 5 = \frac{3}{4}$

$x_3 = \log_{2^3}2^4 = \frac{4}{3}\log_2 2 = \frac{4}{3}$

$x_4 = \log_{2^{1/2}}2^3 = 6\log_2 2 = 6$

then $\sum_{i=1}^{4} x_i = x_1 + x_2 + x_3 + x_4 = \frac{5}{4} + \frac{3}{4} + \frac{4}{3} + 6 = \frac{28}{3}$

Q.11 Let $A = \left(\frac{1}{9}\right)^{\log_3 7}$, $B = 2^{\log_{1/2}7}$, $C = 8^{\frac{-1}{\log_3 2}}$, $D = 4^{-\log_2 6}$. Find $\frac{1}{A} + \frac{1}{B} - \frac{1}{C} - \frac{1}{D}$.

Sol. $A = \left(3^{-2}\right)^{\log_3 7} = 3^{\log_3(7)^{-2}} = \frac{1}{49}$

$B = 2^{\log_{(2^{-1})}7} = 2^{\log_2 7^{-1}} = \frac{1}{7}$

$C = 2^{3(-\log_2 3)} = 2^{\log_2 3^{-3}} = \frac{1}{27}$

$D = 2^{-2\log_2 6} = 2^{\log_2 6^{-2}} = \frac{1}{36}$

$\frac{1}{A} + \frac{1}{B} - \frac{1}{C} - \frac{1}{D} = 49 + 7 - 27 - 36 = \mathbf{-7}$

Q.12 If $p^{\log_3 7} = 81$, then find value of $p^{(\log_3 7)^2}$.

Sol. $p^{\log_3 7 \cdot \log_3 7} = \left(p^{\log_3 7}\right)^{\log_3 7} = (81)^{\log_3 7}$

$= 3^{4\log_3 7} = 3^{\log_3 7^4} = 7^4 = \mathbf{2401}$

Q.13 If a, b, c are real positive numbers such that $a^{\log_3 7} = 27$, $b^{\log_7 11} = 49$, $c^{\log_{11}25} = \sqrt{11}$, then find the value of $a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11}25)^2}$.

(A) 343 (B) 121 (C) 469 (D) 569

Sol. **(C)**

$a^{\log_3 7 \cdot \log_3 7} + b^{\log_7 11 \cdot \log_7 11} + c^{\log_{11}25 \cdot \log_{11}25}$

$= \left(a^{\log_3 7}\right)^{\log_3 7} + \left(b^{\log_7 11}\right)^{\log_7 11} + \left(c^{\log_{11}25}\right)^{\log_{11}25}$

$= (27)^{\log_3 7} + (49)^{\log_7 11} + \left(\sqrt{11}\right)^{\log_{11}25}$

$= 3^{3\log_3 7} + 7^{2\log_7 11} + 11^{\frac{1}{2}\log_{11} 25}$

$= 3^{\log_3 7^3} + 7^{\log_7 11^2} + 11^{\log_{11}(25)^{1/2}}$

$= 7^3 + 11^2 + 5$

$= 343 + 121 + 5 = \mathbf{469}$

Q.14 If $\log_a x = b$ for permissible values of a and x then identify the statement(s) which can be correct?

(A) If a and b are two irrational numbers then x can be rational.

(B) If a rational and b irrational then x can be rational.

(C) If a irrational and b rational then x can be rational.

(D) If a rational and b rational then x can be rational.

Sol. **(ABCD)**

Change this into exponential form

(A) $x = a^b$

Suppose $a = \sqrt{2}$, $b = \log_{\sqrt{2}} 3$

First we will prove that $\log_2 3$ is not rational

Suppose $\log_2 3$ is rational, then $\log_2 3 = p/q \Rightarrow 3 = 2^{p/q}$

$\Rightarrow 3^q = 2^p$

It is not possible for any value of p, q except p = 0 and q = 0

So, $\log_2 3$ is irrational.

Hence $\frac{1}{2}\log_2 3$ is irrational

So, we can say $\log_{\sqrt{2}} 3$ will be irrational

Then $\left(\sqrt{2}\right)^{\log_{\sqrt{2}} 3} = 3$ is rational.

(B) $a = 2$, $b = \log_2 3$

Then $2^{\log_2 3} = 3$ is rational.

(C) $a = \sqrt{3}$, $b = 2$

$\left(\sqrt{3}\right)^2 = 3$ is rational.

(D) $a = 2$, $b = 3$

Then $(2)^3 = 8$ is rational.

Q.15 $B = \frac{1}{\log_3 2} + \frac{2}{\log_9 4} - \frac{3}{\log_{27} 8}$, $C = \frac{\log_3 12}{\log_{36} 3} - \frac{\log_3 4}{\log_{108} 3}$. Find B + C.

(A) 3 (B) 2 (C) 4 (D) 1

Sol. **(B)**

$B = \log_2 3 + 2\log_4 9 - 3\log_8 27$

$= \log_2 3 + 2\log_{2^2} 3^2 - 3\log_{2^3} 3^3$

$= \log_2 3 + 2\log_2 3 - 3\log_2 3 = 0$

$$C = \frac{\log_3 12}{\log_{36} 3} - \frac{\log_3 4}{\log_{108} 3}$$

$C = \log_3 12 \log_3 36 - \log_3 4.\log_3 108$

$C = \log_3(4\times3).\log_3(4\times9) - \log_3 4.\log_3(4\times27)$

$C = (\log_3 4 + \log_3 3)(\log_3 4 + \log_3 9) - \log_3 4(\log_3 4 + \log_3 27)$

$C = (\log_3 4 + 1)(\log_3 4 + 2) - \log_3 4.(\log_3 4 + 3)$

Put $\log_3 4 = t$,

$C = (t + 1)(t + 2) - t(t + 3) = t^2 + 3t + 2 - t^2 - 3t = 2$

So, **B + C = 0 + 2 = 2**

Q.16 $$A = \frac{1}{1+\log_b a+\log_b c} + \frac{1}{1+\log_c a+\log_c b} + \frac{1}{1+\log_a b+\log_a c}$$

Where $a > 0$, $a \neq 1$, $b > 0$, $b \neq 1$, $c > 0$, $c \neq 1$, $abc \neq 1$, then A is :

(A) abc (B) $\frac{1}{abc}$ (C) 1 (D) 0

Sol. **(C)**

$$\frac{1}{\log_b b+\log_b a+\log_b c} + \frac{1}{\log_c c+\log_c a+\log_c b} + \frac{1}{\log_a a+\log_a b+\log_a c}$$

$$= \frac{1}{\log_b abc} + \frac{1}{\log_c abc} + \frac{1}{\log_a abc}$$

$= \log_{abc} b + \log_{abc} c + \log_{abc} a$

$= \log_{abc} abc = 1$

Q.17 Let $a = \log_3 5$, $b = \log_{17} 25$ and $c = \log_5 1000$, $d = \log_7 2058$ then which of the following is/are true?

(A) $a > b$ (B) $a < b$ (C) $c > d$ (D) $c < d$

Sol. **(AC)**

$a = \log_3 5 = \frac{1}{\log_5 3}$ and $b = \log_{17} 25 = \log_{17} 5^2 = 2\log_{17} 5 = \frac{2}{\log_5 17}$

Then $\frac{1}{a} = \log_5 3$ and $\frac{1}{b} = \frac{1}{2}\log_5 17 = \log_5 \sqrt{17}$

From here $\log_5 \sqrt{17} > \log_5 3 \Rightarrow \frac{1}{b} > \frac{1}{a}$

$\Rightarrow$ **a > b**

$c = \log_5 1000$ and $d = \log_7(2058)$

$\because 625 < 1000 < 3125$

$\Rightarrow \log_5 625 < \log_5 1000 < \log_5 3125$

$\Rightarrow \log_5 5^4 < c < \log_5 5^5$

$\Rightarrow 4 < c < 5$

$343 < 2058 < 2401$

$\Rightarrow \quad \log_7 343 < \log_7 2058 < \log_7 2401$

$\Rightarrow \quad \log_7 7^3 < d < \log_7 7^4$

$\Rightarrow \quad 3 < d < 4$

Hence **c > d**

Q.18 Establish the trichotomy between:

(i) $m = (\log_2 5)^2$ & $n = \log_2 20$

Sol. $m - n = (\log_2 5)^2 - (\log_2 20)$

$= (\log_2 5)^2 - \log_2(5\times 4)$

$= (\log_2 5)^2 - (\log_2 5 + 2\log_2 2)$

$= (\log_2 5)^2 - \log_2 5 - 2$

Let $\log_2 5 = t$

then, $m - n = t^2 - t - 2 = (t-2)(t+1)$

$\log_2 5 > 2 \Rightarrow t > 2$ hence $m - n > 0 \Rightarrow$ **m > n**

Q. **(ii)** $a = \log_\pi 2 + \log_2 \pi$ and $b = 1$

Sol. Suppose $\alpha = \log_\pi 2$

Then $a = \alpha + \frac{1}{\alpha} > 2$ and $b = 1$

So **a > b**

Q. If $\log_6 15 = \alpha$ and $\log_{12} 18 = \beta$, then find $\log_{25} 24$ in terms of α, β.

Sol. $\frac{\log_3 15}{\log_3 6} = \alpha \Rightarrow \frac{\log_3(3\times 5)}{\log_3(2\times 3)} = \alpha \Rightarrow \frac{\log_3 3 + \log_3 5}{\log_3 3 + \log_3 2} = \alpha$

$\Rightarrow \frac{1+\log_3 5}{1+\log_3 2} = \alpha$ ... (i)

Also $\frac{\log_3 18}{\log_3 12} = \beta \Rightarrow \frac{\log_3(9\times 2)}{\log_3(3\times 4)} = \beta \Rightarrow \frac{\log_3 9 + \log_3 2}{\log_3 3 + \log_3 4} = \beta$

$\Rightarrow \frac{2+\log_3 2}{1+2\log_3 2} = \beta$ ... (ii)

Now, $\log_{25} 24 = \frac{\log_3 24}{\log_3 25} = \frac{\log_3(8\times 3)}{\log_3 5^2} = \frac{1+3\log_3 2}{2\log_3 5}$

By equation (ii), $2 + \log_3 2 = \beta + 2\beta\log_3 2$

$\Rightarrow (\log_3 2)(1 - 2\beta) = \beta - 2$

$\Rightarrow \log_3 2 = \frac{\beta - 2}{1 - 2\beta}$

By equation (i), $1 + \log_3 5 = \alpha(1 + \log_3 2) = \alpha\left(1+\frac{\beta-2}{1-2\beta}\right)$

$$\Rightarrow 1 + \log_3 5 = \alpha\left(\frac{1-2\beta+\beta-2}{1-2\beta}\right)=\alpha\left(\frac{-1-\beta}{1-2\beta}\right)$$

$$\Rightarrow \log_3 5 = \frac{-\alpha\beta-\alpha-1+2\beta}{1-2\beta}$$

Then $\log_{24} 25 = \frac{1+3\log_3 2}{2\log_3 5} = \frac{1+3\left(\frac{\beta-2}{1-2\beta}\right)}{2\left(\frac{2\beta-1-\alpha\beta-\alpha}{1-2\beta}\right)}$

$$\Rightarrow \log_{24} 25 = \frac{1-2\beta+3\beta-6}{2(2\beta-1-\alpha\beta-\alpha)} = \frac{\beta-5}{2(2\beta-1-\alpha\beta-\alpha)}$$

Q.20 If $\log_7 12 = a$ and $\log_{12} 24 = b$ then find the value of $\log_{54} 168$ in terms of a, b.

Sol. Given, $a = \log_7 12 = \frac{\log_2 12}{\log_2 7} = \frac{\log_2(4\times3)}{\log_2 7} = \frac{2+\log_2 3}{\log_2 7}$ ... (i)

and $b = \log_{12} 24 = \frac{\log_2 24}{\log_2 12} = \frac{\log_2(8\times3)}{\log_2(4\times3)} = \frac{3+\log_2 3}{2+\log_2 3}$ ... (ii)

$\log_{54} 168 = \frac{\log_2 168}{\log_2 54} = \frac{\log_2(7\times3\times8)}{\log_2(27\times2)} = \frac{3+\log_2 7+\log_2 3}{1+3\log_2 3}$ ... (iii)

From equation (ii), $3 + \log_2 3 = 2b + b\log_2 3$

$\Rightarrow (1-b)\log_2 3 = 2b - 3$

$\Rightarrow \log_2 3 = \frac{2b-3}{1-b}$

Replace this in equation (i),

$$\frac{2+\frac{2b-3}{1-b}}{a} = \log_2 7$$

$$\Rightarrow \frac{2-2b+2b-3}{a(1-b)} = \log_2 7$$

$$\Rightarrow \log_2 7 = \frac{-1}{a(1-b)}$$

Then from equation (iii), $\log_{54} 168 = \frac{3+\frac{2b-3}{1-b}-\frac{1}{a(1-b)}}{1+3\left(\frac{2b-3}{1-b}\right)}$

$$\Rightarrow \log_{54} 168 = \frac{3a(1-b)+a(2b-3)-1}{a[(1-b)+3(2b-3)]}$$

$$\Rightarrow \log_{54}168 = \frac{3a - 3ab + 2ab - 3a - 1}{a[1 - b + 6b - 9]}$$

$$\Rightarrow \log_{54}168 = \frac{-1 - ab}{a(5b - 8)}$$

Q.21 If $\log_7\left(\log_7\sqrt{7\sqrt{7\sqrt{7}}}\right) = 1 - a\log_7 2$ and $\log_{15}\left(\log_{15}\sqrt{15\sqrt{15\sqrt{15\sqrt{15}}}}\right) = 1 - b\log_{15}2$.

Find a + b.

Sol. Let $x = \log_7\left(\log_7\sqrt{7\sqrt{7\sqrt{7}}}\right)$

$$x = \log_7\left(\frac{1}{2}\log_7 7\sqrt{7\sqrt{7}}\right)$$

$$x = \log_7\left(\frac{1}{2}\left(\log_7 7 + \log_7\sqrt{7\sqrt{7}}\right)\right)$$

$$x = \log_7\left(\frac{1}{2}\left(1 + \frac{1}{2}\log_7 7\sqrt{7}\right)\right)$$

$$x = \log_7\left(\frac{1}{2}\left(1 + \frac{1}{2}\left(\log_7 7 + \frac{1}{2}\log_7 7\right)\right)\right)$$

$$x = \log_7\left(\frac{1}{2}\left(1 + \frac{1}{2}\left(1 + \frac{1}{2}\right)\right)\right)$$

$$x = \log_7\left(\frac{1}{2}\left(1 + \frac{1}{2}\cdot\frac{3}{2}\right)\right)$$

$$x = \log_7\left(\frac{1}{2}\left(\frac{7}{4}\right)\right) = \log_7\frac{7}{8}$$

$x = 1 - \log_7 8$

$x = 1 - 3\log_7 2$

Compare it with $x = 1 - a\log_7 2$ then $a = 3$

Similarly, suppose $y = \log_{15}\left(\log_{15}\sqrt{15\sqrt{15\sqrt{15\sqrt{15}}}}\right)$

then $y = \log_{15}\left(\frac{1}{2}\log_{15}15\sqrt{15\sqrt{15\sqrt{15}}}\right)$

$$y = \log_{15}\left(\frac{1}{2}\left(1 + \frac{1}{2}\log_{15}15\sqrt{15\sqrt{15}}\right)\right)$$

$$y = \log_{15}\left[\frac{1}{2}\left(1 + \frac{1}{2}\left(1 + \frac{1}{2}\log_{15}15\sqrt{15}\right)\right)\right]$$

$$y = \log_{15}\left[\frac{1}{2}\left(1+\frac{1}{2}\left(1+\frac{1}{2}\left(1+\frac{1}{2}\right)\right)\right)\right]$$

$$y = \log_{15}\left[\frac{1}{2}\left(1+\frac{1}{2}\left(1+\frac{1}{2}\cdot\frac{3}{2}\right)\right)\right]$$

$$y = \log_{15}\left[\frac{1}{2}\left(1+\frac{7}{8}\right)\right] = \log_{15}\left[\frac{1}{2}\cdot\frac{15}{8}\right]$$

$y = \log_{15}15 - \log_{15}2^4$

$y = 1 - 4\log_{15}2$, compare it with $y = 1 - b\log_{15}2$

Hence $b = 4$ and $a = 3$

So $a + b = 4 + 3$

$a + b = 7$

Q.22 If $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$, show that $a^a.b^b.c^c = 1$.

Sol. Assume, $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$

then $\log a = k(b-c) \Rightarrow a\log a = k(ab - ac)$ ... (i)

$\log b = k(c-a) \Rightarrow b\log b = k(bc - ab)$ ... (ii)

$\log c = k(a-b) \Rightarrow c\log c = k(ac - bc)$ ... (iii)

Adding (i), (ii) and (iii), we get

$\Rightarrow \log(a^a.b^b.c^c) = k[ab - ac + bc - ab + ac - bc]$

$\Rightarrow \log(a^a.b^b.c^c) = 0$ (change it into exponential form)

$\Rightarrow$ **$a^a.b^b.c^c = 1$**

Q.23 If a, b, c are positive real numbers other than unity such that $\frac{a(b+c-a)}{\log a} = \frac{b(c+a-b)}{\log b} = \frac{c(a+b-c)}{\log c}$, prove that $a^ba^a = b^cc^b = c^aa^c$.

Sol. Assume,

$$\frac{a(b+c-a)}{\log a} = \frac{b(c+a-b)}{\log b} = \frac{c(a+b-c)}{\log c} = k$$

$$\Rightarrow \log a = \frac{a(b+c-a)}{k} \quad ...(i)$$

$$\log b = \frac{b(c+a-b)}{k} \quad ...(ii)$$

$$\log c = \frac{c(a+b-c)}{k} \quad ...(iii)$$

$$\Rightarrow b\log a = \frac{ab(b+c-a)}{k},\quad a\log b = \frac{ab(c+a-b)}{k},\quad \log c = \frac{c(a+b-c)}{k}$$

$$\Rightarrow \log a^b = \frac{ab(b+c-a)}{k} \quad \ldots(iv)$$

$$\log b^a = \frac{ab(c+a-b)}{k} \quad \ldots(v)$$

Add equations (iv) & (v)

$$\log (a^b.b^a) = \frac{ab}{k}[b + c - a + c + a - b] = \frac{2abc}{k}$$

Consider the base as 'e',

we get, $a^b b^a = e^{\frac{2abc}{k}}$ ... (A)

Now multiply (ii) by c and (iii) by b, we get

$$c \log b = \frac{bc(c+a-b)}{k} \quad \ldots (vi)$$

and $$b \log c = \frac{bc(a+b-c)}{k} \quad \ldots(vii)$$

Add (vi) and (vii), we get

$$\log(b^c.c^b) = \frac{bc}{k}[c + a - b + a + b - c] = \frac{2abc}{k}$$

Change this into exponential form, we get

$$b^c.c^b = e^{\frac{2abc}{k}} \quad \ldots (B)$$

Similarly we get, $c^a.a^c = e^{\frac{2abc}{k}}$ ... (C)

From equations (A), (B) and (C), we get

$$a^b.b^a = b^c.c^b = c^a.a^c$$

Logarithmic Equation

Q.1 Find 'x' in following :

(i) $x^2 + 7^{\log_7 x} - 2 = 0$

Sol. Using $a^{\log_a N} = N$, we get

$x^2 + x - 2 = 0$ and $x > 0$

$\Rightarrow x^2 + 2x - x - 2 = 0$

$\Rightarrow x(x + 2) - 1(x + 2) = 0$

$\Rightarrow (x + 2)(x - 1) = 0$

$\Rightarrow$ either $x = 1$ or $x = -2$

Since $x > 0$, so we get **$x = 1$**

Q. (ii) $2^{\log_2(x^2)} - 3x - 4 = 0$

Sol. Using property $a^{\log_a N} = N$

We get $x^2 - 3x - 4 = 0$

$\Rightarrow$ $x^2 - 4x + x - 4 = 0$

$\Rightarrow$ $x(x-4) + 1(x-4) = 0$

$\Rightarrow$ $(x-4)(x+1) = 0$

$\Rightarrow$ **$x = 4$ and $x = -1$**

Q. (iii) $\dfrac{\log_2(9 - 2^x)}{3 - x} = 1$

Sol. $\log_2(9-2^x) = 3 - x$ **(change it into exponential form)**

$\Rightarrow$ $9 - 2^x = 2^{3-x}$

$\Rightarrow$ $9 - 2^x = 2^3.2^{-x} = \dfrac{8}{2^x}$

$\Rightarrow$ Put $2^x = t$, we get

$9 - t = \dfrac{8}{t}$

$\Rightarrow$ $9t - t^2 = 8$

$\Rightarrow$ $t^2 - 9t + 8 = 0$

$\Rightarrow$ $t^2 - 8t - t + 8 = 0$

$\Rightarrow$ $t(t-8) - 1(t-8) = 0$

$\Rightarrow$ $t = 8, t = 1$

$\Rightarrow$ $2^x = 2^3$ and $2^x = 1$

$\Rightarrow$ $x = 3$ and $x = 0$, but $x \neq 3$, hence **$x = 0$**

Q. (iv) $(x+1)^{\log_{10}(x+1)} = 100(x+1)$

Sol. By taking logarithm both sides with base 10, we get

$\log_{10}(x+1)^{\log_{10}(x+1)} = \log_{10}100(x+1)$

$\Rightarrow$ $\log_{10}(x+1).\log_{10}(x+1) = \log_{10}100 + \log_{10}(x+1)$

Consider, $\log_{10}(x+1) = t$, we get

$t^2 = 2 + t$

$\Rightarrow$ $t^2 - t - 2 = 0$

$\Rightarrow$ $t^2 - 2t + t - 2 = 0$

$\Rightarrow$ $t(t-2) + 1(t-2) = 0$

$\Rightarrow$ $(t-2)(t+1) = 0$

$\Rightarrow$ $t = 2, t = -1$

$\log_{10}(x+1) = 2$ and $\log_{10}(x+1) = -1$

$\Rightarrow$ $x + 1 = 100$ and $(x+1) = \dfrac{1}{10}$

$\Rightarrow$ $x = 100 - 1 = 99$ and $x = \frac{1}{10} - 1$

$\Rightarrow$ $x = 99$ and $x = \frac{-9}{10}$

$\Rightarrow$ **$x = 99$ and $x = -\frac{9}{10}$**

Q. **(v)** $\log_{x-1} 4 = 1 + \log_2 (x-1)$

Sol. $\log_{x-1} 2^2 = 1 + \log_2 (x-1)$

$\Rightarrow$ $\frac{2}{\log_2 (x-1)} = 1 + \log_2 (x-1)$

Consider $\log_2 (x-1) = t$, we get $\frac{2}{t} = 1 + t$

$\Rightarrow$ $2 = t + t^2$

$\Rightarrow$ $t^2 + t - 2 = 0$

$\Rightarrow$ $t^2 + 2t - t - 2 = 0$

$\Rightarrow$ $t(t+2) - 1(t+2) = 0$

$\Rightarrow$ $(t+2)(t-1) = 0$

$\Rightarrow$ $t = -2, t = 1$

$\Rightarrow$ $\log_2 (x-1) = -2$ and $\log_2 (x-1) = 1$

$\Rightarrow$ $(x-1) = 2^{-2}$ and $(x-1) = 2^1$

$\Rightarrow$ $x = 1 + \frac{1}{4}$ and $x = 1 + 2 = 3$

$\Rightarrow$ **$x = \frac{5}{4}$ and $x = 3$**

Q. **(vi)** $1 - \log 5 = \frac{1}{3}\left[\log \frac{1}{2} + \log x + \frac{1}{3}\log 5\right]$

Sol. Base is given as 10

$\log 10 - \log 5 = \frac{1}{3}\left[\log \frac{1}{2} + \log x + \frac{1}{3}\log 5\right]$

$\Rightarrow$ $\log \frac{10}{5} = \frac{1}{3}\left[-\log 2 + \log x + \log 5^{1/3}\right]$

$\Rightarrow$ $3 \log 2 + \log 2 = \log x + \log 5^{1/3}$

$\Rightarrow$ $4 \log 2 - \log 5^{1/3} = \log x$

$\Rightarrow$ $\log x = \log \frac{16}{5^{1/3}}$

$\Rightarrow$ **$x = \frac{16}{5^{1/3}}$**

Q. **(vii)** $3^{\log_3^2 x} + x^{(\log_3 x)} = 162$

Sol. $\left(3^{\log_3 x}\right)^{\log_3 x} + x^{\log_3 x} - 162 = 0$

$\Rightarrow x^{\log_3 x} + x^{\log_3 x} - 162 = 0$

$\Rightarrow 2x^{\log_3 x} = 162$

$\Rightarrow x^{\log_3 x} = 81$

Take logarithm on both sides with base 3, we get

$\log_3 x . \log_3 x = \log_3 81 = 4$

$\Rightarrow (\log_3 x)^2 = 4$

$\Rightarrow \log_3 x = \pm 2$

$\Rightarrow x = 3^{\pm 2}$

$\Rightarrow x = 9, \frac{1}{9}$

Q. **(viii)** $5^{1+\log_4 x} + 5^{(\log_{1/4} x)-1} = \frac{26}{5}$

Sol. $5^{1+\log_4 x} + 5^{-1-\log_4 x} = \frac{26}{5}$

Put $5^{1+\log_4 x} = t$, we get

$$t + \frac{1}{t} = \frac{26}{5}$$

$\Rightarrow t^2 + 1 = \frac{26t}{5}$

$\Rightarrow 5t^2 + 5 = 26t$

$\Rightarrow 5t^2 - 26t + 5 = 0$

$\Rightarrow 5t^2 - 25t - t + 5 = 0$

$\Rightarrow 5t(t-5) - 1(t-5) = 0$

$\Rightarrow (t-5)(5t-1) = 0$

$\Rightarrow t = 5$ or $t = 1/5$

$\Rightarrow 5^{1+\log_4 x} = 5^1$ or $5^{1+\log_4 x} = \frac{1}{5} = 5^{-1}$

$\Rightarrow 1 + \log_4 x = 1$ or $1 + \log_4 x = -1$

$\Rightarrow \log_4 x = 0$ or $\log_4 x = -2$

$\Rightarrow x = 1$ or $x = 4^{-2}$

$\Rightarrow$ **$x = 1$ or $x = \frac{1}{16}$**

Q. **(ix)** $\log_4(2\log_3(1 + \log_2(1 + 3\log_2 x))) = \frac{1}{2}$

Sol. Change it into exponential form,
$2\log_3(1 + \log_2(1 + 3\log_2 x)) = 4^{1/2} = 2$
Again, $1 + \log_2(1 + 3\log_2 x) = 3$
$\log_2(1 + 3\log_2 x) = 2$
Again, $1 + 3\log_2 x = 4$
$\Rightarrow 3\log_2 x = 3$
$\Rightarrow \log_2 x = 1$
$\Rightarrow$ **x = 2**

Q. **(x)** $\log_5\left(5^{1/x} + 125\right) = \log_5 6 + 1 + \frac{1}{2x}$

Sol. $\log_5(5^{1/x} + 125) = \log_5 6 + \left(1 + \frac{1}{2x}\right)\log_5 5$

$\Rightarrow \log_5(5^{1/x} + 125) = \log_5 6 + \log_5 5^{\left(1+\frac{1}{2x}\right)}$
$\Rightarrow \log_5(5^{1/x} + 125) = \log_5(6\times5\times5^{1/2x})$
$\Rightarrow 5^{1/x} + 125 = 30\times5^{1/2x}$
Put $5^{1/2x} = t$, then $t^2 + 125 = 30t$
$\Rightarrow t^2 - 30t + 125 = 0$
$\Rightarrow t^2 - 25t - 5t + 125 = 0$
$\Rightarrow t(t - 25) - 5(t - 25) = 0$
$\Rightarrow (t-25)(t - 5) = 0$
Either t = 25 or t = 5
$\Rightarrow 5^{1/2x} = 5^2$ or $5^{1/2x} = 5^1$
$\Rightarrow \frac{1}{2x} = 2$ or $\frac{1}{2x} = 1$
$\Rightarrow$ **$x = \frac{1}{4}$ or $x = \frac{1}{2}$**

Q. **(xi)** $x^{2+\log_x 4} = x^2 + 27$

Sol. $x^2.x^{\log_x 4} = x^2 + 27$
$\Rightarrow x^2.4 = x^2 + 27$
$\Rightarrow 3x^2 = 27$
$\Rightarrow x^2 = 9$
$\Rightarrow x = +3$ or $x = -3$
As $x > 0$ and $x \neq 1$, so we get **x = 3**

Q. **(xii)** $a^{2\log_2 x} = 5 + 4x^{\log_2 a}$

Sol. $\left(a^{\log_2 x}\right)^2 = 5 + 4\left(a^{\log_2 x}\right)$ (Using property $a^{\log_c b} = b^{\log_c a}$)

Put $a^{\log_2 x} = t$, we get

$t^2 = 5 + 4t$

$\Rightarrow t^2 - 4t - 5 = 0$

$\Rightarrow t^2 - 5t + t - 5 = 0$

$\Rightarrow t(t-5) + 1(t-5) = 0$

$\Rightarrow (t-5)(t+1) = 0$

$\Rightarrow t = 5$ or $t = -1$

$\Rightarrow a^{\log_2 x} = 5$ or $a^{\log_2 x} = -1$ (not possible)

$\Rightarrow \log_5 a.\log_2 x = \log_5 5$

$\Rightarrow \log_2 x.\log_5 a = 1$

$\Rightarrow \log_2 x = \dfrac{1}{\log_5 a}$

$\Rightarrow \log_2 x = \log_a 5$

$\Rightarrow \mathbf{x = 2^{\log_a 5}}$ **or** $\mathbf{5^{\log_a 2}}$

Common logarithm and Natural logarithm:

$\log_{10} N$ is referred as a common logarithm. $\log_e N$ is called as a natural logarithm. It is also written as $\ln N$.

Note:

e is an irrational quantity lying between 2.7 to 2.8. Also $\mathbf{e^{\ln x} = x}$

Characteristics and Mantissa

Real number = Integer + Fraction

$\log_a N$ = Characteristics + Mantissa

$R = I + f$

where R is a real number
I is an integer
f is a fraction i.e. $0 \le f < 1$

Here, characteristics is an integer and mantissa is fraction, so

$0 \le$ mantissa < 1

Ex. (i) $3.4 = 3 + 0.4$
$R = 3.4$, $I = 3$ and $f = 0.4$

(ii) $5.1 = 5 + 0.1$
$R = 5.1$, $I = 5$ and $f = 0.1$

(iii) $-1.2 = -1 - 0.2$
$-1.2 = -1 - 1 + 1 - 0.2$
$-1.2 = -2 + 0.8$
$R = -1.2$, $I = -2$, $f = 0.8$

(iv) $-3.8 = -3 - 0.8$
$-3.8 = -3 - 1 + 1 - 0.8$
$-3.8 = -4 + 0.2$
$R = 3.8$, $I = -4$, $f = 0.2$

(v) $-6.2 = -7 + 0.8$
$R = -6.2$, $I = -7$, $f = 0.8$

Note:

(i) Characteristic can be zero, positive and negative.
(ii) Mantissa is always non-negative and less than 1.

Finding Number of digits

Number of digits = Characteristics + 1

Observation:

Range	Taking log	Characteristics	Number of digits
$1 \le N < 10$	$0 \le \log_{10} N < 1$	0	1
$10 \le N < 100$	$1 \le \log_{10} N < 2$	1	2
$100 \le N < 1000$	$2 \le \log_{10} N < 3$	2	3

By observation, we can say that
Number of digits = Characteristics + 1

Remember the following values

$\log_{10} 2 = 0.3010$
$\log_{10} 3 = 0.4771$
$\log_{10} 5 = 0.6989$
$\log_{10} 7 = 0.8450$

Q.1 Find number of digits in
(i) $N = 6^{50}$

Sol. Taking logarithm on both sides, we get
$\log_{10} N = \log_{10} 6^{50} = 50 \log_{10} 6$

$= 50[\log_{10}2 + \log_{10}3]$
$= 50[0.3010 + 0.4771]$
$= 50[0.7781] = 38.905$

$\log_{10}6^{50} = 38 + 0.905$

Characteristics = 38, So **Number of digits = 39**

Q. **(ii)** $N = 5^{25}$

Sol. Taking logarithm on both sides, we get

$\log_{10}N = \log_{10}5^{25} = 25\log_{10}\frac{10}{2} = 25[\log_{10}10 - \log_{10}2]$

$\log_{10}N = 25[1-0.3010]$
$= 25[0.699]$

$\log_{10}5^{25} = 17.475 = 17 + 0.475$

Here characteristics = 17

Hence, **Number of digits = 18**

Number of zeroes after decimal before a significant digit starts:

Number of zeroes = |Characteristics + 1|

Observation:

Range	Taking log	Characteristics	Number of zeroes
$0.1 \le N < 1$	$-1 \le \log_{10}N < 0$	-1	0
$0.01 \le N < 0.1$	$-2 \le \log_{10}N < -1$	-2	1
$0.001 \le N < 0.01$	$-3 \le \log_{10}N < -2$	-3	2

Hence by observation we can say that
Number of zeros = |Characteristics + 1|

Q.2 Find number of zeroes after decimal point before a significant digit starts in
(i) $N = 3^{-50}$

Sol. Taking logarithm on both sides, we get

$\log_{10}N = \log_{10}3^{-50} = (-50)\log_{10}3$

$\Rightarrow \log_{10}N = (-50) \times (0.4771) = -23.855$

$\Rightarrow \log_{10}N = -23 - 0.855 + 1 - 1 = -24 + 0.145$

Here characteristics $= -24$

Hence **Number of zeros $= |-24 + 1| = |-23| = 23$**

Q. **(ii)** $N = \left(\frac{9}{8}\right)^{-100}$

Sol. Taking logarithm on both sides with base '10'

$$\log_{10}N = \log_{10}\left(\frac{9}{8}\right)^{-100} = -100\log_{10}\left(\frac{9}{8}\right) = -100[2\log 3 - 3\log 2]$$

$\Rightarrow \log_{10}N = -100[2 \times 0.4771 - 3 \times 0.3010]$

$\Rightarrow \log_{10}N = -100[0.9542 - 0.9030]$

$\Rightarrow \log_{10}N = -5.12$

Characteristics $= -6$

Number of zeros $= |-6 + 1| = |-5| = 5$

Q.3 Find 'N' for which characteristics in 2 and base is 10

(A) $N \in [1000, 10000)$ (B) $N \in [10, 100)$

(C) $N \in [100, 1000)$ (D) $N \in [1, 10)$

Sol. **(C)**

According to given condition,

$\Rightarrow 2 \le \log_{10}N < 3$

$\Rightarrow 100 \le N < 1000$

$\Rightarrow$ **$N \in [100, 1000)$**

Q.4 Find the number of integral values of N, for which characteristic is 5 under base 3.

Sol. According to given condition,

$5 \le \log_3 N < 6$

$\Rightarrow 3^5 \le N < 3^6$

$\Rightarrow 243 \le N < 729$

Number of integral values of $N = 729 - 243 =$ **486**

Q.5 Find 'N' for which characteristics is -5 under base 3.

Sol. According to given statement,

$-5 \le \log_3 N < -4$

$\Rightarrow 3^{-5} \le N < 3^{-4}$

$$\Rightarrow \frac{1}{3^5} \le N < \frac{1}{3^4}$$

$$\Rightarrow \frac{1}{243} \le N < \frac{1}{81}$$

$$\Rightarrow N \in \left[\frac{1}{243}, \frac{1}{81}\right)$$

Significance of Modulus function in Logarithmic Equation:

Q.1 Solve : $\log_4 x^2 = 2$

Sol. $x^2 = 4^2$

$x^2 = 16$

$x = \pm 4$

Wrong method:

$\log_4 x^2 = 2$

$\Rightarrow 2\log_4 x = 2$

$\Rightarrow \log_4 x = 1$

$\Rightarrow x = 4^1$

$\Rightarrow x = 4$

Incomplete solution

That's why

$\log(x^2) = 2\log|x|$

Now it can be solved as

$\log_4 x^2 = 2 \Rightarrow 2\log_4|x| = 2$

$\Rightarrow \log_4|x| = 1$

$\Rightarrow |x| = 4^1$

$\Rightarrow$ **$x = \pm 4$**

Note:

(i) $\log x^2 = 2\log|x|$

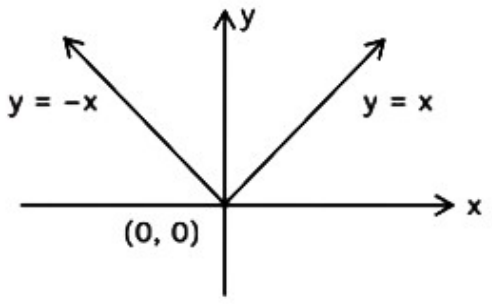
(ii) $\log x^{2m} = 2m \log|x|$, $m \in N$

(iii) $\log x^{2m+1} = (2m+1)\log x$, $m \in N$

(iv) $\sqrt{x^2} = |x|$

Absolute value function/modulus function

$$Y = \sqrt{x^2} = |x| = \begin{cases} -x & ; \ x < 0 \\ 0 & ; \ x = 0 \\ x & ; \ x > 0 \end{cases}$$



Q.2 Solve : $|x-5| = 10$. Find 'x'

Sol.

$x - 5 = \pm 10$

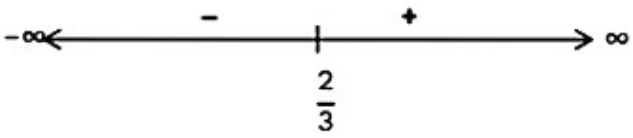
$\Rightarrow$ $x - 5 = 10$ or $x - 5 = -10$

$\Rightarrow$ **$x = 15$ or $x = -5$**

Q.3 Find 'x' if $|3x - 2| + x = 11$

Sol.

Case-(i) When $x < \frac{2}{3}$



Then, $-(3x - 2) + x = 11$

$\Rightarrow$ $-3x + 2 + x = 11$

$\Rightarrow$ $-2x = 9$

$\Rightarrow$ $x = -9/2$

Case-(ii) When $x \geq \frac{2}{3}$

$(3x - 2) + x = 11$

$\Rightarrow$ $4x = 13$

$\Rightarrow$ $x = 13/4$

Hence, **$x = -9/2$ and $x = 13/4$** are the solution.

Q.4 Find 'x' if $|x - 3| + 2|x + 1| = 4$

Sol.



Case-(i) When $x < -1$, then

$-(x - 3) - 2(x + 1) = 4$

$\Rightarrow$ $-x + 3 - 2x - 2 = 4$

$\Rightarrow$ $-3x = 3$

$\Rightarrow$ $x = -1$, but $x < -1$ hence no solution.

Case-(ii) When $-1 \leq x < 3$, then

$-(x - 3) + 2(x + 1) = 4$

$\Rightarrow$ $-x + 3 + 2x + 2 = 4$

$\Rightarrow$ $x = -1$ is a solution.

Case-(iii) When $x \geq 3$, then

$(x - 3) + 2(x + 1) = 4$

$\Rightarrow$ $x - 3 + 2x + 2 = 4$

$\Rightarrow$ $3x = 5$

$\Rightarrow$ $x = \frac{5}{3}$ but $x \geq 3$

Hence no solution

So, **$x = -1$** is the only solution.

Q.5 Find 'x'

(i) $2\log_8(2x) + \log_8(x^2 + 1 - 2x) = \frac{4}{3}$

Sol. $2\log_8(2x) + \log_8(x-1)^2 = \frac{4}{3}$

$\Rightarrow 2\log_8(2x) + 2\log_8|x-1| = \frac{4}{3}$ **(Change it into exponential form)**

$\Rightarrow \log_8 2x|x-1| = \frac{2}{3}$

$\Rightarrow 2x|x-1| = (8)^{\frac{2}{3}} = 4$

$\Rightarrow x|x-1| = 2$

Case (i): When $x < 1$, then

$-x(x-1) = 2$

$\Rightarrow x^2 - x + 2 = 0$

It's discriminant $D = (-1)^2 - 4 \times 2 = 1 - 8 = -7 < 0$

Hence, no real roots exist, so no solution.

Case (ii): When $x \geq 1$, then

$x(x-1) = 2$

$\Rightarrow x^2 - x = 2$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow (x-2)(x+1) = 0$

$\Rightarrow x = -1$ or $x = 2$, but $x \geq 1$

So, **$x = 2$** is the only solution.

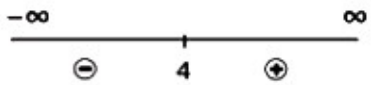
Q. **(ii)** $2\log_3(x-2) + \log_3(x-4)^2 = 0$

Sol. $2\log_3(x-2) + 2\log_3|x-4| = 0$

$\Rightarrow \log_3(x-2)|x-4| = 0$ **(Change it into exponential form)**

$\Rightarrow (x-2)|x-4| = 1$

Case (i): When $x < 4$, then



$-(x-2)(x-4) = 1$

$\Rightarrow x^2 - 6x + 8 = -1$

$\Rightarrow x^2 - 6x + 9 = 0$

$\Rightarrow (x-3)^2 = 0 \Rightarrow x = 3$

Case (ii): When $x \geq 4$

$$(x-2)(x-4)=1$$
$$\Rightarrow x^2-6x+8=1$$
$$\Rightarrow x^2-6x+7=0$$
$$\Rightarrow x=\frac{6\pm\sqrt{8}}{2}$$
$$\Rightarrow x=3+\sqrt{2},\quad x=3-\sqrt{2}$$

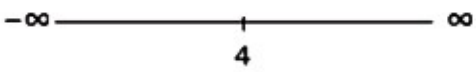
but $x \geq 4$ So, $x=3+\sqrt{2}$

Hence, **$x=3$ or $x=3+\sqrt{2}$** is the solution.

Q. **(iii)** $\log_4(x^2-1)-\log_4(x-1)^2=\log_4\sqrt{(4-x)^2}$

Sol. $\log_4(x^2-1)-\log_4(x-1)^2=\log_4|4-x|$

$$\Rightarrow \log_4\frac{(x^2-1)}{(x-1)^2}=\log_4|4-x|=\log_4|x-4|$$
$$\Rightarrow \frac{(x^2-1)}{(x-1)^2}=|x-4|$$
$$\Rightarrow \frac{(x-1)(x+1)}{(x-1)^2}=|x-4|$$
$$\Rightarrow \frac{(x+1)}{(x-1)}=|x-4|$$



Case (i): When $x<4$

$$\frac{x+1}{x-1}=-(x-4)$$
$$\Rightarrow -x-1=(x-1)(x-4)$$
$$\Rightarrow -x-1=x^2-5x+4$$
$$\Rightarrow x^2-4x+5=0, \text{ Since } D<0$$

Hence no real roots.

Case(ii): When $x \geq 4$

$$\frac{x+1}{x-1}=(x-4)$$

$\Rightarrow (x+1) = (x-4)(x-1)$

$\Rightarrow (x+1) = x^2 - 5x + 4$

$\Rightarrow x - 6x + 3 = 0$

$\Rightarrow x = 3 \pm \sqrt{6}$

$\Rightarrow x = 3+\sqrt{6}, 3-\sqrt{6}$

Since $x \geq 4$ so $\mathbf{x = 3+\sqrt{6}}$

Q. **(iv)** $|x-2|^{10x^2-1} = |x-2|^{3x}$

Sol. Comparing the powers,

$10x^2 - 1 = 3x$

$\Rightarrow 10x^2 - 3x - 1 = 0$

$\Rightarrow 10x^2 - 5x + 2x - 1 = 0$

$5x(2x-1) + (2x-1) = 0$

$\Rightarrow (2x-1)(5x+1) = 0$

$\Rightarrow x = \frac{1}{2}$ or $x = -\frac{1}{5}$

When base is equal to 1, we get

$|x-2| = 1$

$\Rightarrow x - 2 = \pm 1$

$\Rightarrow$ **x = 3 and x = 1**

Q. **(v)** $|x-3|^{3x^2-10x+3} = 1$

Sol. $|x-3|^{3x^2-10x+3} = |x-3|^0$ **(Comparing the powers)**

$\Rightarrow 3x^2 - 10x + 3 = 0$

$\Rightarrow 3x^2 - 9x - x + 3 = 0$

$\Rightarrow 3x(x-3) - 1(x-3) = 0$

$\Rightarrow (x-3)(3x-1) = 0$

$\Rightarrow x = \frac{1}{3}, 3$

When base is equal to 1,

$|x-3| = 1$

$\Rightarrow x - 3 = \pm 1$

$\Rightarrow x = 4, 2$

When x = 3, base is 0, so we reject this.

So **x = 4, 2,** $\frac{1}{3}$ is the solution.

Graphs of logarithm:

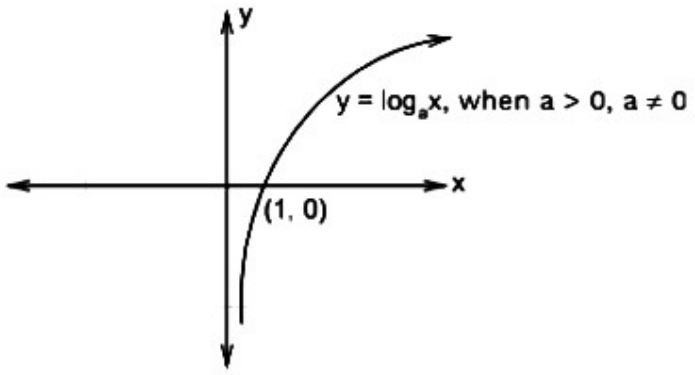
$f(x) = \log_a x, \quad x > 0, \quad a \neq 1, \quad a > 0$

Case-1:

$a > 1$

Ex.: $f(x) = \log_2 x = y$

$x = \frac{1}{2}$,	$y = -1$
$x = \frac{1}{4}$,	$y = -2$
$x = 1$,	$y = 0$
$x = 2$,	$y = 1$
$x = 4$,	$y = 2$
$x = 8$,	$y = 3$
$x = 16$,	$y = 4$

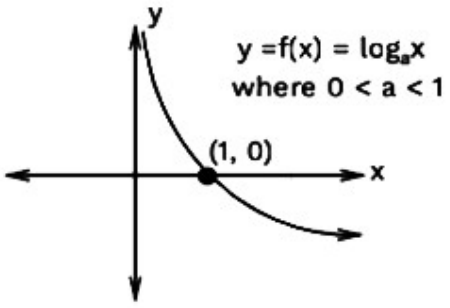


Case-2:

$0 < a < 1$

Ex.: $f(x) = \log_{\frac{1}{2}} x = y$

$x = 2$,	$y = -1$
$x = 4$,	$y = -2$
$x = 8$,	$y = -3$
$x = 16$,	$y = -4$



Note:

1. Whenever the number and base are on the same side of unity then logarithm of that number to the same base is positive.
2. Whenever the number and base are on the opposite side of unity then logarithm of that number to the base is negative.

(i) $y = f(x) = \log_a x > 0 \Rightarrow x,\ a$ are on the same side of unity.

(ii) $y = f(x) = \log_a x < 0 \Rightarrow x,\ a$ are on the opposite side of unity.

Q.1 Solve $\log_4(\log_3 x) + \log_{\frac{1}{4}}\left(\log_{\frac{1}{3}} y\right) = 0$ and $x^2 + y^2 = \frac{17}{4}$.

Sol. $\log_4(\log_3 x) + \log_{\frac{1}{4}}\left(\log_{\frac{1}{3}} y\right) = 0$ .... (i)

$x^2 + y^2 = \frac{17}{4}$ ....(ii)

$\log_4(\log_3 x) - \log_4\left(\log_{\frac{1}{3}} y\right) = 0$

$\Rightarrow \log_4(\log_3 x) = \log_4\left(\log_{\frac{1}{3}} y\right)$

$\log_3 x = \log_{\frac{1}{3}} y$

$\log_3 x = -\log_3 y$

$\log_3 x = \log_3 \frac{1}{y}$

$x = \frac{1}{y}$

Equation (ii), $x^2 + \frac{1}{x^2} = \frac{17}{4}$, put $x^2 = t$

$t + \frac{1}{t} = \frac{17}{4}$

$\Rightarrow \frac{t^2 + 1}{t} = \frac{17}{4}$

$\Rightarrow 4t^2 + 4 = 17t$

$\Rightarrow 4t^2 - 17t + 4 = 0$

$\Rightarrow t = 4, \frac{1}{4}$

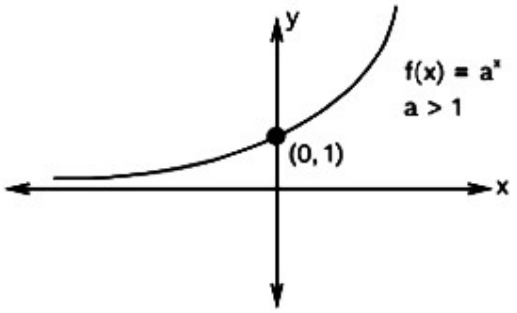
$\Rightarrow x^2 = 4, \frac{1}{4}$

$\Rightarrow x = \pm 2, \pm\frac{1}{2}$ but $x > 0 \Rightarrow x = 2$ and $y = \frac{1}{2}$

Graphs of Exponential

$f(x) = a^x \quad a > 0, \quad a \neq 1$

Case-1: $a > 1$



Ex.: $y = f(x) = 2^x$

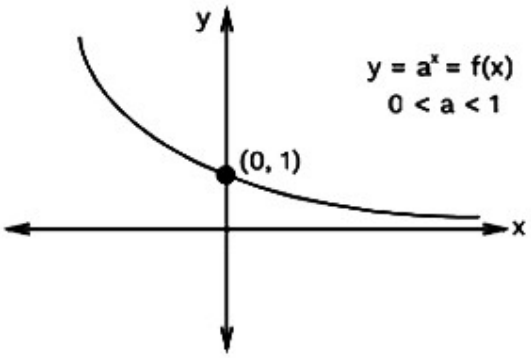
$x = 1, \quad y = 2$

$x = 2, \quad y = 4$

$x = 3, \quad y = 8$

↓ **Increasing**

Case-2: $0 < a < 1$



Ex.: $f(x) = \left(\frac{1}{2}\right)^x$

$x = 1, \quad y = \frac{1}{2}$

$x = 2, \quad y = \frac{1}{4}$

$x = 3, \quad y = \frac{1}{8}$

↓ **Decreasing**

$x \to \infty \Rightarrow f(x) \to 0$

$x \to -\infty \Rightarrow f(x) \to \infty$

Logarithmic Inequalities

1.

$$f(x) = \log_a x$$

(i) When $a > 1$ and $x_1 > x_2 \Leftrightarrow \log_a x_1 > \log_a x_2$

(ii) When $0 < a < 1$ and $x_1 > x_2 \Leftrightarrow \log_a x_1 < \log_a x_2$

2.

$$f(x) = a^x$$

(i) When $a > 1$ and $x_1 > x_2 \Leftrightarrow a^{x_1} > a^{x_2}$

(ii) When $0 < a < 1$ and $x_1 > x_2 \Leftrightarrow a^{x_1} < a^{x_2}$

Q.1 Find 'x'

(i) $\log_{\frac{1}{2}}(2x+1) > 0$

Sol. $\log_{\frac{1}{2}}(2x+1) > \log_{\frac{1}{2}} 1$

$\Rightarrow (2x+1) < 1$

$\Rightarrow x < 0$

and $(2x+1) > 0$

$\Rightarrow x > -\frac{1}{2}$

By Intersection, $x \in \left(-\frac{1}{2}, 0\right)$

Q. **(ii)** $\log_2(x+2) > 0$

Sol. $\log_2(x+2) > \log_2 1$

$\Rightarrow (x + 2) > 1$

$\Rightarrow x > -1$

and $x + 2 > 0$

$\Rightarrow x > -2$

By intersection, we get **$x > -1$ or $x \in (-1, \infty)$**.

Q. (iii) $2^{3-6x} > 1$

Sol. $2^{3-6x} > 2^0$

$\Rightarrow 3 - 6x > 0$

$\Rightarrow 6x < 3$

$\Rightarrow x < \frac{3}{6}$

$\Rightarrow x < \frac{1}{2}$

$x \in \left(-\infty, \frac{1}{2}\right)$

Q. (iv) $\left(\frac{1}{5}\right)^{2x+1} > 125$

Sol. $5^{-(2x+1)} > 5^3$

$\Rightarrow -2x - 1 > 3$

$\Rightarrow -2x > 4$

$\Rightarrow x < -2$

$\Rightarrow x \in (-\infty, -2)$

Q. (v) $\log_{0.3}|x-2| > 0$

Sol. $\log_{0.3}|x-2| > \log_{0.3} 1$

$\Rightarrow |x - 2| < 1$

$\Rightarrow -1 < x - 2 < 1$

$\Rightarrow 1 < x < 3$

and $|x - 2| = 1 \Rightarrow x - 2 = \pm 1 \Rightarrow x = 3$ or 1

and $|x - 2| \neq 0 \Rightarrow x \neq 2$

By Intersection, $x \in (1,2) \cup (2,3)$ or $x \in (1,3) - \{2\}$

Q. (vi) $\log_8(x^2 - 4x + 3) \leq 1$

Sol. $\log_8(x^2 - 4x + 3) \leq \log_8 8$

$\Rightarrow x^2 - 4x + 3 \leq 8$ and $x^2 - 4x + 3 > 0$

$\Rightarrow x^2 - 4x - 5 \leq 0$ and $x^2 - 4x + 3 > 0$

$\Rightarrow x^2 - 5x + x - 5 \leq 0$ and $x^2 - 3x - x + 3 > 0$

$\Rightarrow x(x-5)+1(x-5)\leq 0$ and $x(x-3)-1(x-3)>0$

$\Rightarrow (x-5)(x+1)\leq 0$ and $(x-3)(x-1)>0$

$\Rightarrow x\in[-1,5]$ and $x\in(-\infty,1)\cup(3,\infty)$

By intersection, $x\in[-1,1)\cup(3,5]$

Q. **(vii)** $\log_{\frac{1}{2}}\left(\log_6\left(\frac{x^2+x}{x+4}\right)\right)<0$

Sol. (i) $\log_6\left(\frac{x^2+x}{x+4}\right)>\left(\frac{1}{2}\right)^0$

$\frac{x^2+x}{x+4}>6$

(ii) $\log_6\left(\frac{x^2+x}{x+4}\right)>0$

$\frac{x^2+x}{x+4}>1$

(iii) $\frac{x^2+x}{x+4}>0$

By intersection of (i), (ii) and (iii) we can clearly state that

$\frac{x^2+x}{x+4}>6$

$\Rightarrow \frac{x^2+x}{x+4}-6>0$

$\Rightarrow \frac{x^2+x-6x-24}{x+4}>0$

$\Rightarrow \frac{x^2-5x-24}{(x+4)}>0$

$\Rightarrow \frac{(x-8)(x+3)}{(x+4)}>0$

$\Rightarrow x\in(-4,-3)\cup(8,\infty)$

Q. (viii) $(0.3)^{\log_{\frac{1}{3}}\left(\log_2\left(\frac{3x+6}{x^2+2}\right)\right)} > 1$

Sol. $(0.3)^{\log_{\frac{1}{3}}\left(\log_2\left(\frac{3x+6}{x^2+2}\right)\right)} > (0.3)^0$ and $\log_2\left(\frac{3x+6}{x^2+2}\right) > 0$ and $\frac{3x+6}{x^2+2} > 0$

So, $\log_{\frac{1}{3}}\left(\log_2\left(\frac{3x+6}{x^2+2}\right)\right) < 0$ and $\frac{3x+6}{x^2+2} > 1$ and $\frac{3x+6}{x^2+2} > 0$

$\Rightarrow \log_2\left(\frac{3x+6}{x^2+2}\right) > 1$

$\Rightarrow \frac{3x+6}{x^2+2} > 2$

By intersection, we get

$\frac{3x+6}{x^2+2} > 2$

$\Rightarrow 3x + 6 > 2(x^2 + 2)$
$\Rightarrow 2x^2 + 4 < 3x + 6$
$\Rightarrow 2x^2 - 3x - 2 < 0$
$\Rightarrow 2x^2 - 4x + x - 2 < 0$
$\Rightarrow 2x(x - 2) + 1(x - 2) < 0$
$\Rightarrow (x - 2)(2x + 1) < 0$

$\Rightarrow x \in \left(-\frac{1}{2}, 2\right)$

Q.2 Find the solution of the equation, $2\log_9\left(2\left(\frac{1}{2}\right)^x - 1\right) = \log_{27}\left(\left(\frac{1}{4}\right)^x - 4\right)^3$. Also state whether the solution is rational or irrational.

Sol. Suppose $\left(\frac{1}{2}\right)^x = t$ so $\left(\frac{1}{4}\right)^x = t^2$

$$\frac{2}{2}\log_3(2t - 1) = \frac{3}{3}\log_3(t^2 - 4)$$

$\Rightarrow 2t - 1 = t^2 - 4$
$\Rightarrow t^2 - 2t - 3 = 0$
$\Rightarrow (t - 3)(t + 1) = 0$
$\Rightarrow t = 3, t = -1$

$\Rightarrow \left(\frac{1}{2}\right)^x = 3$ or $\left(\frac{1}{2}\right)^x = -1$ **(Not possible)**

$\Rightarrow 2^{-x} = 3$

$\Rightarrow -x \log 2 = \log 3$

$\Rightarrow x = -\dfrac{\log 3}{\log 2}$

$\Rightarrow$ **$x = -\log_2(3)$ which is irrational.**

Q.3 If the product of all solutions of equation $\dfrac{(2020)x}{2021} = (2020)^{\log_x 2021}$ can be expressed in the lowest form as $\dfrac{m}{n}.\ (m,\ n \in I)$ then find the value of (m + n).

Sol. $\log_{2021}\left(\dfrac{2020x}{2021}\right) = \log_x(2021)\cdot\log_{(2021)}(2020)$

$\Rightarrow \log_{2021}2020 + \log_{2021}x - 1 = \log_x 2021\cdot\log_{2021}2020$

Suppose $\log_{2021}x = t$ then $\log_x 2021 = \dfrac{1}{t}$

Then, $\log_{2021}2020 + t - 1 = \dfrac{1}{t}\log_{2021}2020$

$t\cdot\log_{2021}2020 + t^2 - t - \log_{2021}2020 = 0$

$\Rightarrow (t-1)\log_{2021}(2020) + t(t-1) = 0$

$\Rightarrow (t-1)(\log_{2021}2020 + t) = 0$

$\Rightarrow t = 1$ or $t = -\log_{2021}2020$

$\Rightarrow \log_{2021}x = 1$ or $\log_{2021}x = \log_{2021}(2020)^{-1}$

$\Rightarrow x = 2021$ or $x = \dfrac{1}{2020}$

Product of solutions

$$\frac{m}{n} = \frac{2021}{2020}$$

$m + n = 2021 + 2020 = 4041$

Q.4 The sum of the integral values(s) of $a \in [-9, 9]$ so that equation $(x-2)\log_{13}3 + \log_{13}(3^x - 7a) = \log_{13}2 + 2\log_{13}a$ has integral solution.

Sol. $\log_{13}3^{x-2}\cdot(3^x - 7a) = \log_{13}(2\cdot a^2)$

$\Rightarrow 3^{x-2}(3^x - 7a) = 2a^2$

Put $3^x = t$, we get

$\dfrac{t}{9}(t - 7a) = 2a^2$

$\Rightarrow t^2 - 7at = 18a^2$

$\Rightarrow t^2 - 9at + 2at - 18a^2 = 0$

$\Rightarrow (t - 9a)(t + 2a) = 0$

$\Rightarrow t = 9a$ or $t = -2a$

$\Rightarrow 3^x = 9a$ or $3^x = -2a$

$3^0, 3^1, 3^2, 3^3, \ldots = -2a$ or

$1, 3, 9, 27, \ldots = -2a$ or

No integral value of a

$3^0, 3^1, 3^2, \ldots = 9a$

$1, 3, 9, \ldots = 9a$

$a = \frac{1}{9}, \frac{3}{9}, \frac{9}{9}, \ldots$

$a = \frac{1}{9}, \frac{1}{3}, 1, 3, 9, \ldots$

$a = 1, 3, 9$

Sum = 1 + 3 + 9 = 13

Q.5 For i = 1 to 6, let $\log_a(\log_b(\log_c x_i))=0$, where a, b and c represent every possible different arrangement of 2, 4 and 8. The product $x_1x_2x_3x_4x_5x_6$ can be expressed in the form 2^N. The value of N, is

(A) 20 (B) 28 (C) 33 (D) 50

Sol. (D)

$\log_a(\log_b(\log_c x_i)) = 0$

$\Rightarrow \log_b(\log_c x_i) = a^0 = 1$

$\Rightarrow \log_c x_i = b$

$\Rightarrow x_i = c^b$

a = 2	a = 4	a = 8
$x = 4^8, 8^4$	$x = 2^8, 8^2$	$x = 2^4, 4^2$

$x_1x_2x_3x_4x_5x_6 = 8^2 \cdot 4^8 \cdot 8^4 \cdot 2^8 \cdot 4^2 \cdot 2^4 = 2^{16} \cdot 2^{12} \cdot 2^8 \cdot 2^6 \cdot 2^4 \cdot 2^4 = 2^{50}$

N = 50